

Diagram showing the SWD connection for J4 header. The connections are:

- SWCLK (red line) to Pin 1
- SWDIO (green line) to Pin 3
- GND (green line) to Pin 2
- GND (green line) to Pin 4

J2 LEFT

+1V8	1
GND	2
TUN_G0	3
TUN_G1	4
CFG_SDA	5
CFG_SCL	6
GP21	7
GP22	8
GP23	9
GP24	10
GP26	11
GP27	12
GP28	13
GP29	14
+3V3	15
VBUS	16

J3 RIGHT

ST_INT	1
ST_LINK	2
TUN_G2	3
TUN_G3	4
RF_EN_GP	5
GP7	6
GP6	7
GP5	8
GP4	9
GP3	10
GP2	11
GP1	12
GP0	13
GND	14
+3V3	15
VBUS	16

The RP2350A, core regulator, crystal, flash and decoupling are Raspberry Pi's RP2350A minimal design (RP-006440), part for part: the crystal is a 2520 with 27 p where that has an ABM8 and 15 p.

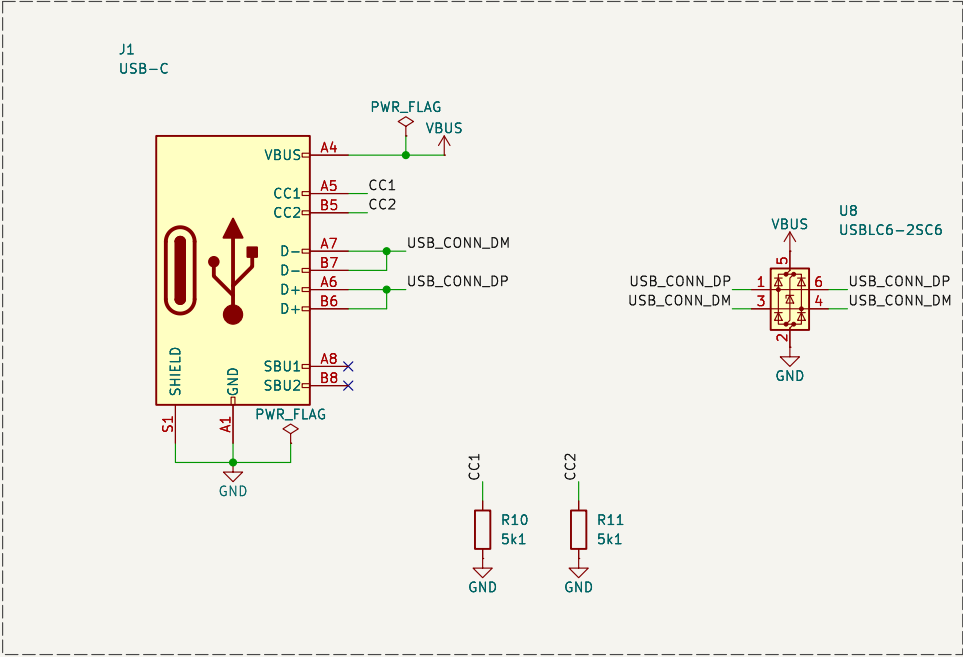
GPIO: 0-7 right-hand header, 8/9 USB path selects. 10 RF_EN, 11 1.8 V enable, 12/13 tunnelled SDA/SCL (I2C0), 14 LINK_STATUS, 15 MODE_INT, 16/17 tunnelled TXD/RXD (UART0), 18/19 configuration bus, 20 transceiver enable, 21-24, 26-29 left-hand header, 25 LED.

The headers carry the 3.3 V side of every line the ST60 takes from a host, so a host that is not the RP2350 can turn the link.

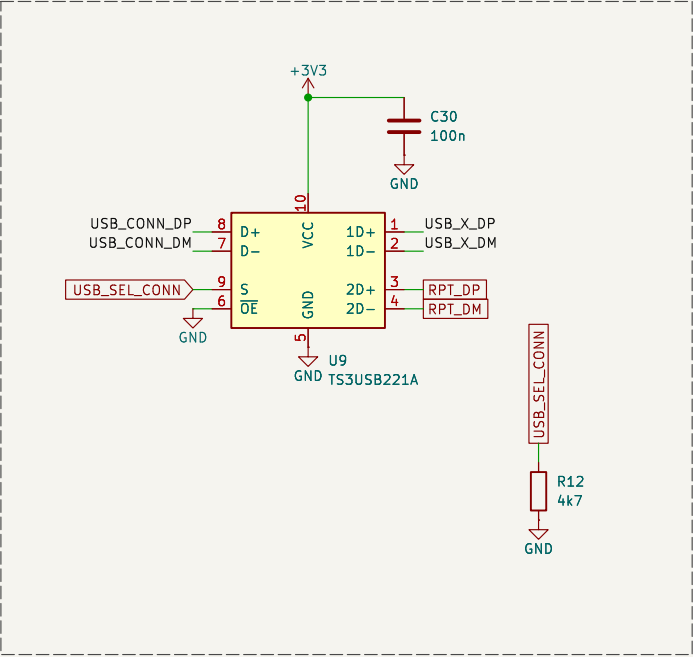
The schematic diagram shows the PTN3222GM transceiver circuit. The chip is powered by +1V8 (VDD1V8) and +3V3 (VDD3V3). It has two 100pF capacitors (C27, C29) and two 470nF capacitors (C26, C28) connected to the power pins. The chip is connected to an I2C interface (RPT_DP, RPT_DM, ST_LINK_STATUS, ST_SCL, ST_SDA) and an SPI interface (RST_N, SCL, SDA, ADDR). The chip is labeled U5 PTN3222GM.

The schematic diagram illustrates the internal and external components of the ST60A3H1 module. The module is represented by a yellow rectangle with various pins and internal components. The internal components include a red resistor R20 (0R) connected to a +1V8 supply, a green diode PWR_FLAG, a green capacitor C21 (100n) and C22 (100n) connected to GND, and a green capacitor C3 connected to VDD_1V8. The external components include a red resistor R24 (4k7) connected to a +1V8 supply, a green capacitor C25 (4k7) connected to a +1V8 supply, a red resistor R26 (4k7) connected to a +1V8 supply, a green capacitor C27 (4k7) connected to a +1V8 supply, a red resistor R21 (1k5) connected to a +1V8 supply, a green capacitor R22 (1k8) connected to GND, and a red resistor R31 (4k7) connected to GND. The module is labeled U4 ST60A3H1.

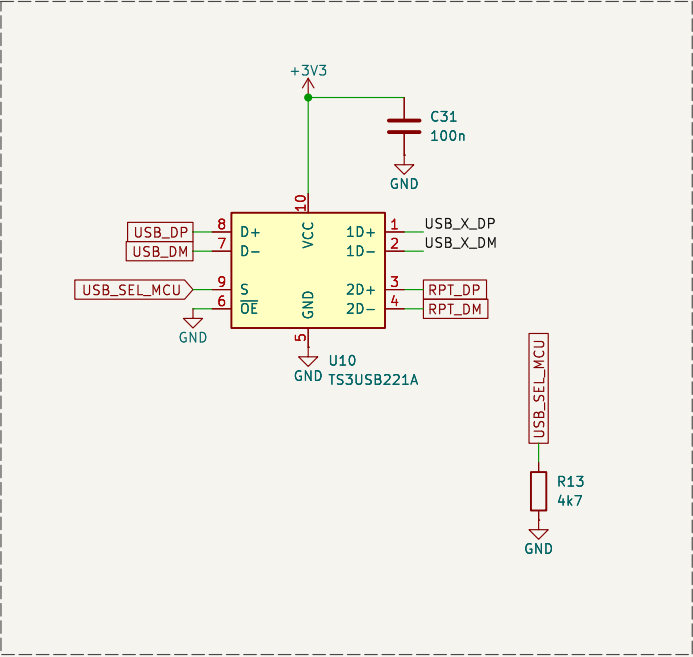
USB-C, device



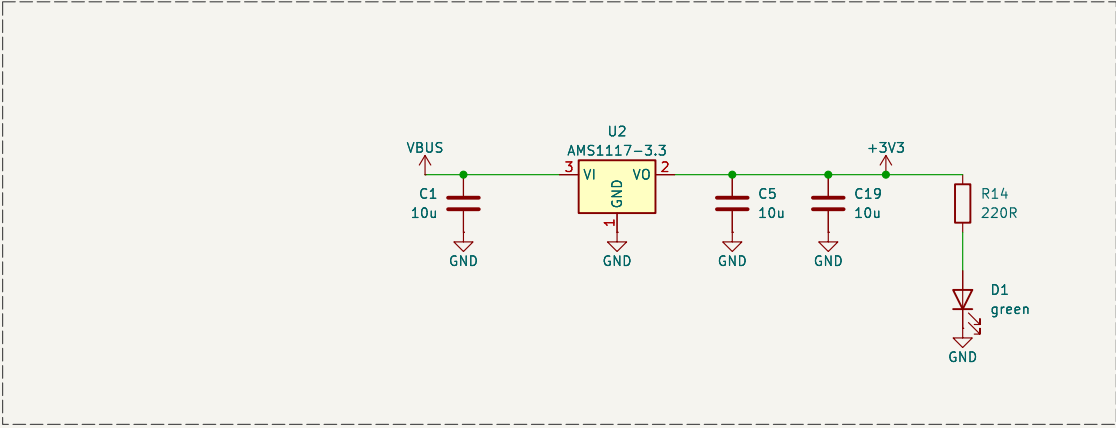
Switch: connector side



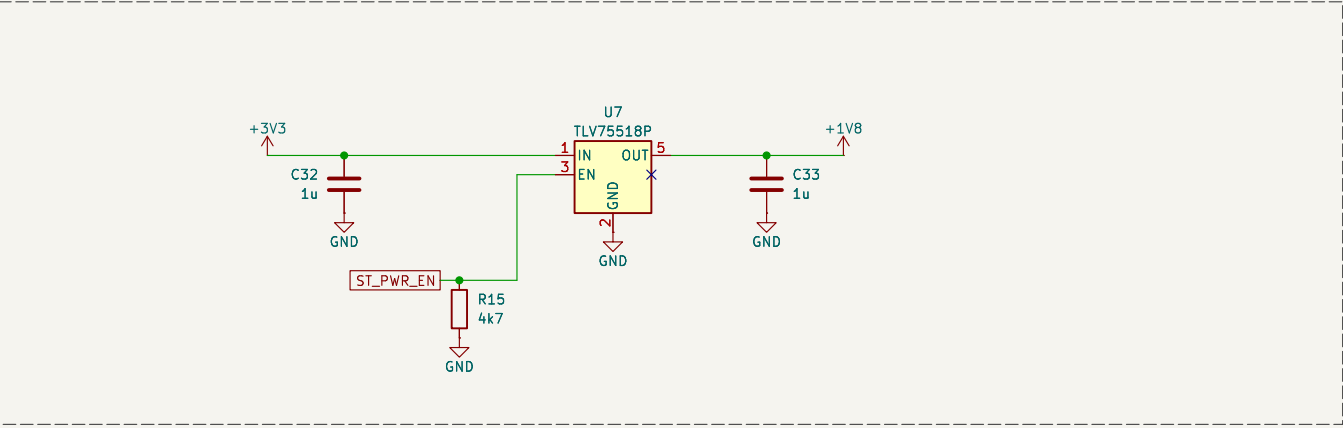
Switch: RP2350 side



3.3 V

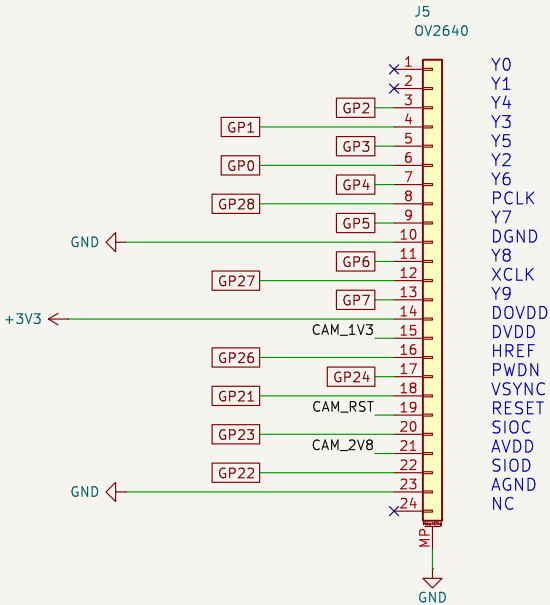


1.8 V, switchable

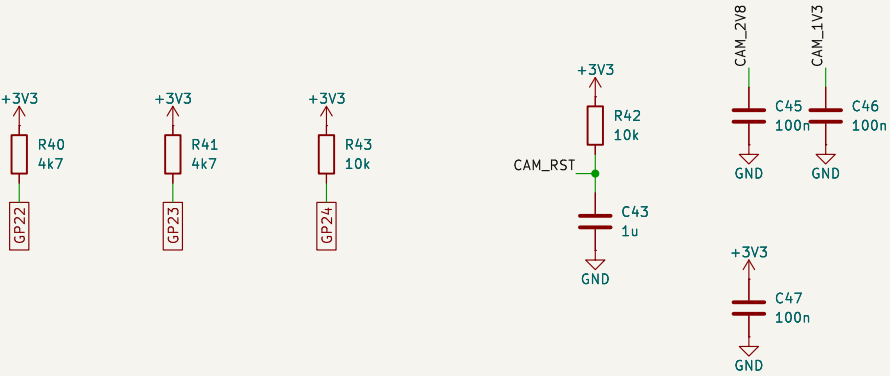


Three things want the USB pair: the connector, the RP2350, the eUSB2 repeater. The two switches join any two through the node USB_X.
USB_SEL_CONN low, USB_SEL_MCU low: connector to RP2350 -- the board as it powers up, both selects pulled down.
CONN high: connector to repeater -- a host's USB goes over the air.
MCU high: RP2350 to repeater -- the RP2350 is the device at the far end.
Both high joins all three and is not a state to use.
ST_PWR_EN high turns the 1.8 V regulator on; it is pulled down, so the rail is off until the RP2350 asks for it. Off, the regulator discharges its output, which is a power-on reset for the ST60A3H1.

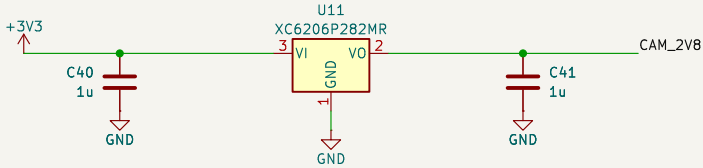
OV2640 module, 24-way flex (on the back)



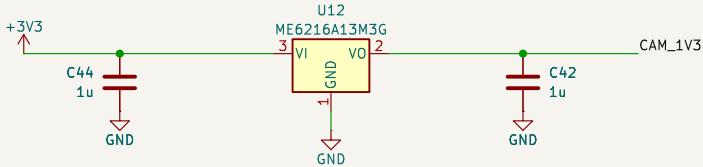
Pull-ups, reset, decoupling at the connector



2.8 V, the sensor's analogue supply



1.3 V, the sensor's core



For the common OV2640 module (24-way 0.5 mm flex, DVP; the one on the ESP32-CAM). The connector is numbered as that board's schematic numbers it; module makers number the tail the other way round (their pin 1 is NC, here 24). The connector is on the back and bottom-contact: the tail goes in with its contacts toward the board, and the lens then looks away from it. Every camera signal is on a header GPIO: 0-7 data, 21 VSYNC, 22/23 SCCB (I2C1), 24 PWDN (pulled up; asleep until driven low), 26 HREF, 27 XCLK (PWM 5B or PIO), 28 PCLK. RESET has no pin: 10k / 1u, and the sensor's software reset.